Chunyang Luo

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Research Interests: Agent Systems | Human-Agent Interaction | LLM-based Planning & Reasoning | Multi-Robot Planning

EDUCATION

B.Eng. Candidate in Robotics Engineering, Peking University 2023 – Present

Expected Graduation: June 2027; GPA: 3.54/4.0 (85/100); Rank: 5/19 (Top 26%)

Selected Core Courses: Foundations of Machine Learning (98), Probability and Statistics (94), Introduction to Robotics (89)

PUBLICATIONS & MANUSCRIPTS

- Baiqiao Zhang, Junze Li, Chunyang Luo, Yuying Tang, Wenshuo Zhang, Shiyuan Piao, Dingdong Liu, Shuai Ma, and Xiaojuan Ma. *MemoSync: Co-Managing Developer-Agent Metamemory Across Contexts.* Submitted to ACM CHI 2027.

RESEARCH EXPERIENCES

OpenGRAIL: Integrating LLM and Classical Planning for Multi-Robot Task Planning in Open Worlds Sept. 2025 – Present

Role: Lead Researcher *Supervisor:* Prof. Zhongkui Li

- Developed an LLM–classical planning framework that translates natural-language instructions into executable multi-robot plans.
- Designed hierarchical task decomposition and MILP-based heterogeneous robot allocation and scheduling under capability, temporal, spatial, and collaboration constraints.
- Conducted preliminary evaluations in household scenarios, demonstrating higher task-execution accuracy and lower makespan and runtime than existing methods.

Human-Agent Collaboration for Reliable Long-Horizon Agent Systems Jul. 2026 – Sept. 2026

Summer Research, Hong Kong University of Science and Technology (HKUST)

Role: Research Assistant *Supervisor:* Prof. Xiaojuan Ma

- Investigated how AI agents maintain task context, user intent, and reusable knowledge during long-horizon interactions.
- Studied existing agent memory mechanisms and contributed to the conceptual design of memory representations and workflow structures for improving agent reliability.
- Designed evaluation scenarios and comparison experiments to analyze the impact of memory organization strategies on task completion, consistency, and knowledge reuse.

Multi-Robot Collaborative LTL Continuous Planning for Dynamic Environments Jul. 2025 – Present

Role: Research Assistant *Supervisor:* Prof. Zhongkui Li

- Designed an online task decomposition and replanning framework for heterogeneous multi-robot LTL tasks under dynamic events.
- Built a heterogeneous multi-robot simulation platform (Unitree Go2, Scout) with task execution and state feedback interfaces.
- Benchmarked against static and periodic-replanning baselines on completion rate, planning time, and cumulative cost; extending to physical robots.

AWARDS & COMPETITIONS

- **Second Prize**, 22nd "Jiang Zehan Cup" Modeling and Computer Applications Competition, Peking University
- **Honorable Mention**, MCM/ICM 2026

SELECTED PROJECTS & TRAINING

RepoFlow: Repository-Level Workflow for AI Coding Agents *(Course Project, Peking University)* Mar. 2026 – Present

Designed persistent-context and structured agent-handoff mechanisms for task continuity in long-horizon coding agents.

Alibaba DAMO Academy – Embodied Intelligence Bootcamp Spring 2026

Built a robotic-arm visual-navigation demo integrating perception, control, and task execution.

SKILLS AND OTHERS

- **Programming & Tools:** Python, MATLAB, Linux, Git, LaTeX
- **Planning & AI:** LTL, PDDL, Classical Planning, Multi-Robot Task Allocation, LLM-based Planning & Reasoning, RL
- **Robotics:** ROS, Gazebo, RViz, AI2-THOR
- **Language:** CET-6 578, IELTS Overall 7.0